Model Architecture: gemma-4-E2B

Generated by sg_modelload

July 21, 2026

1 Model Overview

Model Name	gemma-4-E2B
Path	/models/memorization/gemma-4-E2B
Total Parameters	5,104,297,504
Weight Dtype	bfloat16

2 Forward Pass Algorithm

Algorithm 1 Forward pass of gemma-4-E2B

$h \leftarrow \text{embed_tokens}(x) \in \mathbb{R}^{B \times T \times 8960}$
for $layer \leftarrow 0$ **to** 34 **do**
    $h \leftarrow \text{layer}_{layer}(h)$
end for
$h \leftarrow \text{norm}(h) \in \mathbb{R}^{B \times T \times 8960}$
$\text{logits} \leftarrow \text{lm_head}(h) \in \mathbb{R}^{B \times T \times 262144}$

3 Layer-by-Layer Architecture

All weight tensors and their dimensions are listed below.

Layer Path	Type
model.audio_tower.layers.0	Gemma4AudioLayer
model.audio_tower.layers.0.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.0.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.0.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.0.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.0.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.0.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.0.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.0.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.0.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.0.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.0.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.0.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.0.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.0.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.0.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.0.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.0.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.0.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.0.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.0.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.0.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.0.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm

Layer Path	Type
model.audio_tower.layers.0.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.0.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.0.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.0.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.0.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.0.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.0.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.0.self_attn.post.linear.weight	Linear
model.audio_tower.layers.0.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.0.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.0.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.0.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.0.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.0	Gemma4TextDecoderLayer
model.language_model.layers.0.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.0.mlp	Gemma4TextMLP
model.language_model.layers.0.mlp.down_proj.weight	Linear
model.language_model.layers.0.mlp.gate_proj.weight	Linear
model.language_model.layers.0.mlp.up_proj.weight	Linear
model.language_model.layers.0.per_layer_input_gate.weight	Linear
model.language_model.layers.0.per_layer_projection.weight	Linear
model.language_model.layers.0.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.0.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.0.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.0.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.0.self_attn	Gemma4TextAttention
model.language_model.layers.0.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.0.self_attn.k_proj.weight	Linear
model.language_model.layers.0.self_attn.o_proj.weight	Linear
model.language_model.layers.0.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.0.self_attn.q_proj.weight	Linear
model.language_model.layers.0.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.0	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.0.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.0.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.0.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.0.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.0.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.0.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.0.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.0.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.0.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.0.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.0.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.0.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.0.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.0.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.0.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.0.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.0.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.0.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.0.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.0.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.0.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.0.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.1	Gemma4AudioLayer
model.audio_tower.layers.1.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.1.feed_forward1.ffw_layer_1	Gemma4ClippableLinear

Layer Path	Type
model.audio_tower.layers.1.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.1.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.1.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.1.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.1.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.1.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.1.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.1.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.1.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.1.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.1.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.1.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.1.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.1.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.1.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.1.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.1.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.1.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.1.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.1.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.1.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.1.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.1.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.1.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.1.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.1.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.1.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.1.self_attn.post.linear.weight	Linear
model.audio_tower.layers.1.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.1.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.1.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.1.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.1.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.1	Gemma4TextDecoderLayer
model.language_model.layers.1.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.1.mlp	Gemma4TextMLP
model.language_model.layers.1.mlp.down_proj.weight	Linear
model.language_model.layers.1.mlp.gate_proj.weight	Linear
model.language_model.layers.1.mlp.up_proj.weight	Linear
model.language_model.layers.1.per_layer_input_gate.weight	Linear
model.language_model.layers.1.per_layer_projection.weight	Linear
model.language_model.layers.1.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.1.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.1.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.1.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.1.self_attn	Gemma4TextAttention
model.language_model.layers.1.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.1.self_attn.k_proj.weight	Linear
model.language_model.layers.1.self_attn.o_proj.weight	Linear
model.language_model.layers.1.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.1.self_attn.q_proj.weight	Linear
model.language_model.layers.1.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.1	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.1.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.1.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.1.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.1.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.1.mlp.gate_proj	Gemma4ClippableLinear

Layer Path	Type
model.vision_tower.encoder.layers.1.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.1.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.1.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.1.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.1.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.1.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.1.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.1.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.1.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.1.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.1.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.1.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.1.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.1.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.1.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.1.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.1.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.2	Gemma4AudioLayer
model.audio_tower.layers.2.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.2.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.2.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.2.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.2.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.2.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.2.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.2.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.2.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.2.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.2.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.2.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.2.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.2.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.2.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.2.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.2.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.2.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.2.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.2.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.2.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.2.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.2.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.2.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.2.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.2.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.2.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.2.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.2.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.2.self_attn.post.linear.weight	Linear
model.audio_tower.layers.2.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.2.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.2.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.2.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.2.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.2	Gemma4TextDecoderLayer
model.language_model.layers.2.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.2.mlp	Gemma4TextMLP
model.language_model.layers.2.mlp.down_proj.weight	Linear
model.language_model.layers.2.mlp.gate_proj.weight	Linear

Layer Path	Type
model.language_model.layers.2.mlp.up_proj.weight	Linear
model.language_model.layers.2.per_layer_input_gate.weight	Linear
model.language_model.layers.2.per_layer_projection.weight	Linear
model.language_model.layers.2.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.2.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.2.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.2.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.2.self_attn	Gemma4TextAttention
model.language_model.layers.2.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.2.self_attn.k_proj.weight	Linear
model.language_model.layers.2.self_attn.o_proj.weight	Linear
model.language_model.layers.2.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.2.self_attn.q_proj.weight	Linear
model.language_model.layers.2.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.2	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.2.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.2.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.2.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.2.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.2.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.2.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.2.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.2.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.2.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.2.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.2.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.2.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.2.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.2.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.2.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.2.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.2.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.2.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.2.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.2.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.2.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.2.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.3	Gemma4AudioLayer
model.audio_tower.layers.3.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.3.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.3.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.3.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.3.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.3.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.3.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.3.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.3.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.3.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.3.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.3.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.3.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.3.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.3.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.3.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.3.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.3.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.3.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.3.lconv1d.linear_start	Gemma4ClippableLinear

Layer Path	Type
model.audio_tower.layers.3.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.3.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.3.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.3.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.3.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.3.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.3.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.3.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.3.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.3.self_attn.post.linear.weight	Linear
model.audio_tower.layers.3.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.3.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.3.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.3.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.3.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.3	Gemma4TextDecoderLayer
model.language_model.layers.3.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.3.mlp	Gemma4TextMLP
model.language_model.layers.3.mlp.down_proj.weight	Linear
model.language_model.layers.3.mlp.gate_proj.weight	Linear
model.language_model.layers.3.mlp.up_proj.weight	Linear
model.language_model.layers.3.per_layer_input_gate.weight	Linear
model.language_model.layers.3.per_layer_projection.weight	Linear
model.language_model.layers.3.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.3.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.3.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.3.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.3.self_attn	Gemma4TextAttention
model.language_model.layers.3.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.3.self_attn.k_proj.weight	Linear
model.language_model.layers.3.self_attn.o_proj.weight	Linear
model.language_model.layers.3.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.3.self_attn.q_proj.weight	Linear
model.language_model.layers.3.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.3	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.3.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.3.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.3.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.3.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.3.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.3.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.3.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.3.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.3.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.3.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.3.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.3.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.3.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.3.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.3.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.3.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.3.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.3.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.3.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.3.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.3.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.3.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.4	Gemma4AudioLayer

Layer Path	Type
model.audio_tower.layers.4.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.4.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.4.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.4.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.4.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.4.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.4.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.4.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.4.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.4.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.4.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.4.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.4.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.4.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.4.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.4.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.4.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.4.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.4.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.4.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.4.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.4.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.4.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.4.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.4.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.4.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.4.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.4.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.4.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.4.self_attn.post.linear.weight	Linear
model.audio_tower.layers.4.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.4.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.4.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.4.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.4.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.4	Gemma4TextDecoderLayer
model.language_model.layers.4.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.4.mlp	Gemma4TextMLP
model.language_model.layers.4.mlp.down_proj.weight	Linear
model.language_model.layers.4.mlp.gate_proj.weight	Linear
model.language_model.layers.4.mlp.up_proj.weight	Linear
model.language_model.layers.4.per_layer_input_gate.weight	Linear
model.language_model.layers.4.per_layer_projection.weight	Linear
model.language_model.layers.4.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.4.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.4.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.4.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.4.self_attn	Gemma4TextAttention
model.language_model.layers.4.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.4.self_attn.k_proj.weight	Linear
model.language_model.layers.4.self_attn.o_proj.weight	Linear
model.language_model.layers.4.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.4.self_attn.q_proj.weight	Linear
model.language_model.layers.4.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.4	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.4.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.4.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.4.mlp.down_proj	Gemma4ClippableLinear

Layer Path	Type
model.vision_tower.encoder.layers.4.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.4.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.4.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.4.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.4.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.4.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.4.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.4.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.4.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.4.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.4.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.4.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.4.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.4.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.4.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.4.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.4.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.4.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.4.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.5	Gemma4AudioLayer
model.audio_tower.layers.5.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.5.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.5.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.5.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.5.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.5.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.5.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.5.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.5.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.5.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.5.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.5.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.5.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.5.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.5.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.5.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.5.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.5.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.5.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.5.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.5.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.5.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.5.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.5.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.5.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.5.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.5.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.5.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.5.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.5.self_attn.post.linear.weight	Linear
model.audio_tower.layers.5.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.5.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.5.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.5.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.5.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.5	Gemma4TextDecoderLayer
model.language_model.layers.5.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.5.mlp	Gemma4TextMLP

Layer Path	Type
model.language_model.layers.5.mlp.down_proj.weight	Linear
model.language_model.layers.5.mlp.gate_proj.weight	Linear
model.language_model.layers.5.mlp.up_proj.weight	Linear
model.language_model.layers.5.per_layer_input_gate.weight	Linear
model.language_model.layers.5.per_layer_projection.weight	Linear
model.language_model.layers.5.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.5.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.5.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.5.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.5.self_attn	Gemma4TextAttention
model.language_model.layers.5.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.5.self_attn.k_proj.weight	Linear
model.language_model.layers.5.self_attn.o_proj.weight	Linear
model.language_model.layers.5.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.5.self_attn.q_proj.weight	Linear
model.language_model.layers.5.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.5	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.5.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.5.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.5.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.5.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.5.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.5.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.5.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.5.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.5.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.5.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.5.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.5.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.5.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.5.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.5.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.5.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.5.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.5.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.5.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.5.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.5.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.5.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.6	Gemma4AudioLayer
model.audio_tower.layers.6.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.6.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.6.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.6.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.6.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.6.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.6.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.6.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.6.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.6.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.6.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.6.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.6.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.6.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.6.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.6.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.6.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.6.lconv1d.linear_end	Gemma4ClippableLinear

Layer Path	Type
model.audio_tower.layers.6.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.6.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.6.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.6.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.6.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.6.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.6.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.6.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.6.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.6.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.6.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.6.self_attn.post.linear.weight	Linear
model.audio_tower.layers.6.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.6.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.6.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.6.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.6.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.6	Gemma4TextDecoderLayer
model.language_model.layers.6.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.6.mlp	Gemma4TextMLP
model.language_model.layers.6.mlp.down_proj.weight	Linear
model.language_model.layers.6.mlp.gate_proj.weight	Linear
model.language_model.layers.6.mlp.up_proj.weight	Linear
model.language_model.layers.6.per_layer_input_gate.weight	Linear
model.language_model.layers.6.per_layer_projection.weight	Linear
model.language_model.layers.6.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.6.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.6.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.6.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.6.self_attn	Gemma4TextAttention
model.language_model.layers.6.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.6.self_attn.k_proj.weight	Linear
model.language_model.layers.6.self_attn.o_proj.weight	Linear
model.language_model.layers.6.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.6.self_attn.q_proj.weight	Linear
model.language_model.layers.6.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.6	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.6.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.6.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.6.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.6.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.6.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.6.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.6.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.6.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.6.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.6.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.6.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.6.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.6.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.6.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.6.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.6.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.6.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.6.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.6.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.6.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.6.self_attn.v_proj	Gemma4ClippableLinear

Layer Path	Type
model.vision_tower.encoder.layers.6.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.7	Gemma4AudioLayer
model.audio_tower.layers.7.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.7.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.7.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.7.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.7.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.7.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.7.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.7.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.7.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.7.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.7.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.7.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.7.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.7.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.7.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.7.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.7.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.7.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.7.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.7.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.7.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.7.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.7.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.7.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.7.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.7.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.7.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.7.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.7.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.7.self_attn.post.linear.weight	Linear
model.audio_tower.layers.7.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.7.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.7.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.7.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.7.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.7	Gemma4TextDecoderLayer
model.language_model.layers.7.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.7.mlp	Gemma4TextMLP
model.language_model.layers.7.mlp.down_proj.weight	Linear
model.language_model.layers.7.mlp.gate_proj.weight	Linear
model.language_model.layers.7.mlp.up_proj.weight	Linear
model.language_model.layers.7.per_layer_input_gate.weight	Linear
model.language_model.layers.7.per_layer_projection.weight	Linear
model.language_model.layers.7.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.7.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.7.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.7.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.7.self_attn	Gemma4TextAttention
model.language_model.layers.7.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.7.self_attn.k_proj.weight	Linear
model.language_model.layers.7.self_attn.o_proj.weight	Linear
model.language_model.layers.7.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.7.self_attn.q_proj.weight	Linear
model.language_model.layers.7.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.7	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.7.input_layernorm.weight	Gemma4RMSNorm

Layer Path	Type
model.vision_tower.encoder.layers.7.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.7.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.7.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.7.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.7.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.7.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.7.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.7.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.7.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.7.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.7.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.7.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.7.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.7.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.7.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.7.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.7.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.7.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.7.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.7.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.7.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.8	Gemma4AudioLayer
model.audio_tower.layers.8.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.8.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.8.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.8.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.8.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.8.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.8.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.8.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.8.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.8.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.8.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.8.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.8.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.8.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.8.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.8.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.8.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.8.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.8.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.8.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.8.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.8.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.8.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.8.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.8.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.8.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.8.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.8.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.8.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.8.self_attn.post.linear.weight	Linear
model.audio_tower.layers.8.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.8.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.8.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.8.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.8.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.8	Gemma4TextDecoderLayer

Layer Path	Type
model.language_model.layers.8.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.8.mlp	Gemma4TextMLP
model.language_model.layers.8.mlp.down_proj.weight	Linear
model.language_model.layers.8.mlp.gate_proj.weight	Linear
model.language_model.layers.8.mlp.up_proj.weight	Linear
model.language_model.layers.8.per_layer_input_gate.weight	Linear
model.language_model.layers.8.per_layer_projection.weight	Linear
model.language_model.layers.8.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.8.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.8.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.8.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.8.self_attn	Gemma4TextAttention
model.language_model.layers.8.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.8.self_attn.k_proj.weight	Linear
model.language_model.layers.8.self_attn.o_proj.weight	Linear
model.language_model.layers.8.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.8.self_attn.q_proj.weight	Linear
model.language_model.layers.8.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.8	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.8.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.8.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.8.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.8.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.8.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.8.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.8.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.8.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.8.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.8.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.8.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.8.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.8.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.8.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.8.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.8.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.8.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.8.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.8.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.8.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.8.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.8.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.9	Gemma4AudioLayer
model.audio_tower.layers.9.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.9.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.9.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.9.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.9.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.9.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.9.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.9.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.9.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.9.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.9.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.9.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.9.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.9.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.9.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.9.lconv1d.conv_norm.weight	Gemma4RMSNorm

Layer Path	Type
model.audio_tower.layers.9.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.9.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.9.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.9.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.9.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.9.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.9.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.9.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.9.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.9.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.9.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.9.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.9.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.9.self_attn.post.linear.weight	Linear
model.audio_tower.layers.9.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.9.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.9.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.9.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.9.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.9	Gemma4TextDecoderLayer
model.language_model.layers.9.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.9.mlp	Gemma4TextMLP
model.language_model.layers.9.mlp.down_proj.weight	Linear
model.language_model.layers.9.mlp.gate_proj.weight	Linear
model.language_model.layers.9.mlp.up_proj.weight	Linear
model.language_model.layers.9.per_layer_input_gate.weight	Linear
model.language_model.layers.9.per_layer_projection.weight	Linear
model.language_model.layers.9.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.9.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.9.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.9.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.9.self_attn	Gemma4TextAttention
model.language_model.layers.9.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.9.self_attn.k_proj.weight	Linear
model.language_model.layers.9.self_attn.o_proj.weight	Linear
model.language_model.layers.9.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.9.self_attn.q_proj.weight	Linear
model.language_model.layers.9.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.9	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.9.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.9.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.9.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.9.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.9.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.9.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.9.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.9.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.9.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.9.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.9.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.9.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.9.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.9.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.9.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.9.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.9.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.9.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.9.self_attn.q_proj	Gemma4ClippableLinear

Layer Path	Type
model.vision_tower.encoder.layers.9.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.9.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.9.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.10	Gemma4AudioLayer
model.audio_tower.layers.10.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.10.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.10.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.10.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.10.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.10.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.10.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.10.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.10.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.10.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.10.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.10.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.10.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.10.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.10.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.10.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.10.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.10.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.10.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.10.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.10.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.10.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.10.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.10.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.10.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.10.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.10.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.10.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.10.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.10.self_attn.post.linear.weight	Linear
model.audio_tower.layers.10.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.10.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.10.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.10.self_attn.v_proj	Gemma4ClippableLinear
model.audio_tower.layers.10.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.10	Gemma4TextDecoderLayer
model.language_model.layers.10.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.10.mlp	Gemma4TextMLP
model.language_model.layers.10.mlp.down_proj.weight	Linear
model.language_model.layers.10.mlp.gate_proj.weight	Linear
model.language_model.layers.10.mlp.up_proj.weight	Linear
model.language_model.layers.10.per_layer_input_gate.weight	Linear
model.language_model.layers.10.per_layer_projection.weight	Linear
model.language_model.layers.10.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.10.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.10.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.10.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.10.self_attn	Gemma4TextAttention
model.language_model.layers.10.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.10.self_attn.k_proj.weight	Linear
model.language_model.layers.10.self_attn.o_proj.weight	Linear
model.language_model.layers.10.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.10.self_attn.q_proj.weight	Linear
model.language_model.layers.10.self_attn.v_proj.weight	Linear

Layer Path	Type
model.vision_tower.encoder.layers.10	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.10.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.10.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.10.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.10.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.10.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.10.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.10.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.10.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.10.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.10.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.10.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.10.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.10.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.10.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.10.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.10.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.10.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.10.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.10.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.10.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.10.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.10.self_attn.v_proj.linear.weight	Linear
model.audio_tower.layers.11	Gemma4AudioLayer
model.audio_tower.layers.11.feed_forward1	Gemma4AudioFeedForward
model.audio_tower.layers.11.feed_forward1.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.11.feed_forward1.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.11.feed_forward1.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.11.feed_forward1.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.11.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.11.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.11.feed_forward2	Gemma4AudioFeedForward
model.audio_tower.layers.11.feed_forward2.ffw_layer_1	Gemma4ClippableLinear
model.audio_tower.layers.11.feed_forward2.ffw_layer_1.linear.weight	Linear
model.audio_tower.layers.11.feed_forward2.ffw_layer_2	Gemma4ClippableLinear
model.audio_tower.layers.11.feed_forward2.ffw_layer_2.linear.weight	Linear
model.audio_tower.layers.11.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.11.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.11.lconv1d	Gemma4AudioLightConv1d
model.audio_tower.layers.11.lconv1d.conv_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.11.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d
model.audio_tower.layers.11.lconv1d.linear_end	Gemma4ClippableLinear
model.audio_tower.layers.11.lconv1d.linear_end.linear.weight	Linear
model.audio_tower.layers.11.lconv1d.linear_start	Gemma4ClippableLinear
model.audio_tower.layers.11.lconv1d.linear_start.linear.weight	Linear
model.audio_tower.layers.11.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm
model.audio_tower.layers.11.norm_out.weight	Gemma4RMSNorm
model.audio_tower.layers.11.norm_post_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.11.norm_pre_attn.weight	Gemma4RMSNorm
model.audio_tower.layers.11.self_attn.per_dim_scale	Gemma4AudioAttention
model.audio_tower.layers.11.self_attn.k_proj	Gemma4ClippableLinear
model.audio_tower.layers.11.self_attn.k_proj.linear.weight	Linear
model.audio_tower.layers.11.self_attn.post	Gemma4ClippableLinear
model.audio_tower.layers.11.self_attn.post.linear.weight	Linear
model.audio_tower.layers.11.self_attn.q_proj	Gemma4ClippableLinear
model.audio_tower.layers.11.self_attn.q_proj.linear.weight	Linear
model.audio_tower.layers.11.self_attn.relative_k_proj.weight	Linear
model.audio_tower.layers.11.self_attn.v_proj	Gemma4ClippableLinear

Layer Path	Type
model.audio_tower.layers.11.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.11	Gemma4TextDecoderLayer
model.language_model.layers.11.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.11.mlp	Gemma4TextMLP
model.language_model.layers.11.mlp.down_proj.weight	Linear
model.language_model.layers.11.mlp.gate_proj.weight	Linear
model.language_model.layers.11.mlp.up_proj.weight	Linear
model.language_model.layers.11.per_layer_input_gate.weight	Linear
model.language_model.layers.11.per_layer_projection.weight	Linear
model.language_model.layers.11.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.11.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.11.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.11.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.11.self_attn	Gemma4TextAttention
model.language_model.layers.11.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.11.self_attn.k_proj.weight	Linear
model.language_model.layers.11.self_attn.o_proj.weight	Linear
model.language_model.layers.11.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.11.self_attn.q_proj.weight	Linear
model.language_model.layers.11.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.11	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.11.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.11.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.11.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.11.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.11.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.11.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.11.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.11.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.11.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.11.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.11.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.11.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.11.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.11.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.11.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.11.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.11.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.11.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.11.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.11.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.11.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.11.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.12	Gemma4TextDecoderLayer
model.language_model.layers.12.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.12.mlp	Gemma4TextMLP
model.language_model.layers.12.mlp.down_proj.weight	Linear
model.language_model.layers.12.mlp.gate_proj.weight	Linear
model.language_model.layers.12.mlp.up_proj.weight	Linear
model.language_model.layers.12.per_layer_input_gate.weight	Linear
model.language_model.layers.12.per_layer_projection.weight	Linear
model.language_model.layers.12.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.12.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.12.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.12.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.12.self_attn	Gemma4TextAttention
model.language_model.layers.12.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.12.self_attn.k_proj.weight	Linear

Layer Path	Type
model.language_model.layers.12.self_attn.o_proj.weight	Linear
model.language_model.layers.12.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.12.self_attn.q_proj.weight	Linear
model.language_model.layers.12.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.12	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.12.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.12.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.12.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.12.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.12.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.12.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.12.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.12.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.12.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.12.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.12.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.12.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.12.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.12.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.12.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.12.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.12.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.12.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.12.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.12.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.12.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.12.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.13	Gemma4TextDecoderLayer
model.language_model.layers.13.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.13.mlp	Gemma4TextMLP
model.language_model.layers.13.mlp.down_proj.weight	Linear
model.language_model.layers.13.mlp.gate_proj.weight	Linear
model.language_model.layers.13.mlp.up_proj.weight	Linear
model.language_model.layers.13.per_layer_input_gate.weight	Linear
model.language_model.layers.13.per_layer_projection.weight	Linear
model.language_model.layers.13.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.13.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.13.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.13.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.13.self_attn	Gemma4TextAttention
model.language_model.layers.13.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.13.self_attn.k_proj.weight	Linear
model.language_model.layers.13.self_attn.o_proj.weight	Linear
model.language_model.layers.13.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.13.self_attn.q_proj.weight	Linear
model.language_model.layers.13.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.13	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.13.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.13.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.13.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.13.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.13.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.13.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.13.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.13.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.13.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.13.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.13.pre_feedforward_layernorm.weight	Gemma4RMSNorm

Layer Path	Type
model.vision_tower.encoder.layers.13.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.13.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.13.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.13.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.13.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.13.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.13.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.13.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.13.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.13.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.13.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.14	Gemma4TextDecoderLayer
model.language_model.layers.14.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.14.mlp	Gemma4TextMLP
model.language_model.layers.14.mlp.down_proj.weight	Linear
model.language_model.layers.14.mlp.gate_proj.weight	Linear
model.language_model.layers.14.mlp.up_proj.weight	Linear
model.language_model.layers.14.per_layer_input_gate.weight	Linear
model.language_model.layers.14.per_layer_projection.weight	Linear
model.language_model.layers.14.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.14.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.14.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.14.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.14.self_attn	Gemma4TextAttention
model.language_model.layers.14.self_attn.k_norm.weight	Gemma4RMSNorm
model.language_model.layers.14.self_attn.k_proj.weight	Linear
model.language_model.layers.14.self_attn.o_proj.weight	Linear
model.language_model.layers.14.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.14.self_attn.q_proj.weight	Linear
model.language_model.layers.14.self_attn.v_proj.weight	Linear
model.vision_tower.encoder.layers.14	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.14.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.14.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.14.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.14.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.14.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.14.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.14.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.14.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.14.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.14.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.14.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.14.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.14.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.14.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.14.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.14.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.14.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.14.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.14.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.14.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.14.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.14.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.15	Gemma4TextDecoderLayer
model.language_model.layers.15.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.15.mlp	Gemma4TextMLP
model.language_model.layers.15.mlp.down_proj.weight	Linear
model.language_model.layers.15.mlp.gate_proj.weight	Linear

Layer Path	Type
model.language_model.layers.15.mlp.up_proj.weight	Linear
model.language_model.layers.15.per_layer_input_gate.weight	Linear
model.language_model.layers.15.per_layer_projection.weight	Linear
model.language_model.layers.15.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.15.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.15.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.15.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.15.self_attn	Gemma4TextAttention
model.language_model.layers.15.self_attn.o_proj.weight	Linear
model.language_model.layers.15.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.15.self_attn.q_proj.weight	Linear
model.vision_tower.encoder.layers.15	Gemma4VisionEncoderLayer
model.vision_tower.encoder.layers.15.input_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.15.mlp	Gemma4VisionMLP
model.vision_tower.encoder.layers.15.mlp.down_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.15.mlp.down_proj.linear.weight	Linear
model.vision_tower.encoder.layers.15.mlp.gate_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.15.mlp.gate_proj.linear.weight	Linear
model.vision_tower.encoder.layers.15.mlp.up_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.15.mlp.up_proj.linear.weight	Linear
model.vision_tower.encoder.layers.15.post_attention_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.15.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.15.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.15.self_attn	Gemma4VisionAttention
model.vision_tower.encoder.layers.15.self_attn.k_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.15.self_attn.k_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.15.self_attn.k_proj.linear.weight	Linear
model.vision_tower.encoder.layers.15.self_attn.o_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.15.self_attn.o_proj.linear.weight	Linear
model.vision_tower.encoder.layers.15.self_attn.q_norm.weight	Gemma4RMSNorm
model.vision_tower.encoder.layers.15.self_attn.q_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.15.self_attn.q_proj.linear.weight	Linear
model.vision_tower.encoder.layers.15.self_attn.v_proj	Gemma4ClippableLinear
model.vision_tower.encoder.layers.15.self_attn.v_proj.linear.weight	Linear
model.language_model.layers.16	Gemma4TextDecoderLayer
model.language_model.layers.16.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.16.mlp	Gemma4TextMLP
model.language_model.layers.16.mlp.down_proj.weight	Linear
model.language_model.layers.16.mlp.gate_proj.weight	Linear
model.language_model.layers.16.mlp.up_proj.weight	Linear
model.language_model.layers.16.per_layer_input_gate.weight	Linear
model.language_model.layers.16.per_layer_projection.weight	Linear
model.language_model.layers.16.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.16.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.16.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.16.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.16.self_attn	Gemma4TextAttention
model.language_model.layers.16.self_attn.o_proj.weight	Linear
model.language_model.layers.16.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.16.self_attn.q_proj.weight	Linear
model.language_model.layers.17	Gemma4TextDecoderLayer
model.language_model.layers.17.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.17.mlp	Gemma4TextMLP
model.language_model.layers.17.mlp.down_proj.weight	Linear
model.language_model.layers.17.mlp.gate_proj.weight	Linear
model.language_model.layers.17.mlp.up_proj.weight	Linear
model.language_model.layers.17.per_layer_input_gate.weight	Linear
model.language_model.layers.17.per_layer_projection.weight	Linear

Layer Path	Type
model.language_model.layers.17.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.17.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.17.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.17.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.17.self_attn	Gemma4TextAttention
model.language_model.layers.17.self_attn.o_proj.weight	Linear
model.language_model.layers.17.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.17.self_attn.q_proj.weight	Linear
model.language_model.layers.18	Gemma4TextDecoderLayer
model.language_model.layers.18.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.18.mlp	Gemma4TextMLP
model.language_model.layers.18.mlp.down_proj.weight	Linear
model.language_model.layers.18.mlp.gate_proj.weight	Linear
model.language_model.layers.18.mlp.up_proj.weight	Linear
model.language_model.layers.18.per_layer_input_gate.weight	Linear
model.language_model.layers.18.per_layer_projection.weight	Linear
model.language_model.layers.18.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.18.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.18.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.18.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.18.self_attn	Gemma4TextAttention
model.language_model.layers.18.self_attn.o_proj.weight	Linear
model.language_model.layers.18.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.18.self_attn.q_proj.weight	Linear
model.language_model.layers.19	Gemma4TextDecoderLayer
model.language_model.layers.19.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.19.mlp	Gemma4TextMLP
model.language_model.layers.19.mlp.down_proj.weight	Linear
model.language_model.layers.19.mlp.gate_proj.weight	Linear
model.language_model.layers.19.mlp.up_proj.weight	Linear
model.language_model.layers.19.per_layer_input_gate.weight	Linear
model.language_model.layers.19.per_layer_projection.weight	Linear
model.language_model.layers.19.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.19.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.19.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.19.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.19.self_attn	Gemma4TextAttention
model.language_model.layers.19.self_attn.o_proj.weight	Linear
model.language_model.layers.19.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.19.self_attn.q_proj.weight	Linear
model.language_model.layers.20	Gemma4TextDecoderLayer
model.language_model.layers.20.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.20.mlp	Gemma4TextMLP
model.language_model.layers.20.mlp.down_proj.weight	Linear
model.language_model.layers.20.mlp.gate_proj.weight	Linear
model.language_model.layers.20.mlp.up_proj.weight	Linear
model.language_model.layers.20.per_layer_input_gate.weight	Linear
model.language_model.layers.20.per_layer_projection.weight	Linear
model.language_model.layers.20.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.20.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.20.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.20.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.20.self_attn	Gemma4TextAttention
model.language_model.layers.20.self_attn.o_proj.weight	Linear
model.language_model.layers.20.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.20.self_attn.q_proj.weight	Linear
model.language_model.layers.21	Gemma4TextDecoderLayer
model.language_model.layers.21.input_layernorm.weight	Gemma4RMSNorm

Layer Path	Type
model.language_model.layers.21.mlp	Gemma4TextMLP
model.language_model.layers.21.mlp.down_proj.weight	Linear
model.language_model.layers.21.mlp.gate_proj.weight	Linear
model.language_model.layers.21.mlp.up_proj.weight	Linear
model.language_model.layers.21.per_layer_input_gate.weight	Linear
model.language_model.layers.21.per_layer_projection.weight	Linear
model.language_model.layers.21.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.21.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.21.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.21.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.21.self_attn	Gemma4TextAttention
model.language_model.layers.21.self_attn.o_proj.weight	Linear
model.language_model.layers.21.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.21.self_attn.q_proj.weight	Linear
model.language_model.layers.22	Gemma4TextDecoderLayer
model.language_model.layers.22.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.22.mlp	Gemma4TextMLP
model.language_model.layers.22.mlp.down_proj.weight	Linear
model.language_model.layers.22.mlp.gate_proj.weight	Linear
model.language_model.layers.22.mlp.up_proj.weight	Linear
model.language_model.layers.22.per_layer_input_gate.weight	Linear
model.language_model.layers.22.per_layer_projection.weight	Linear
model.language_model.layers.22.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.22.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.22.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.22.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.22.self_attn	Gemma4TextAttention
model.language_model.layers.22.self_attn.o_proj.weight	Linear
model.language_model.layers.22.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.22.self_attn.q_proj.weight	Linear
model.language_model.layers.23	Gemma4TextDecoderLayer
model.language_model.layers.23.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.23.mlp	Gemma4TextMLP
model.language_model.layers.23.mlp.down_proj.weight	Linear
model.language_model.layers.23.mlp.gate_proj.weight	Linear
model.language_model.layers.23.mlp.up_proj.weight	Linear
model.language_model.layers.23.per_layer_input_gate.weight	Linear
model.language_model.layers.23.per_layer_projection.weight	Linear
model.language_model.layers.23.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.23.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.23.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.23.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.23.self_attn	Gemma4TextAttention
model.language_model.layers.23.self_attn.o_proj.weight	Linear
model.language_model.layers.23.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.23.self_attn.q_proj.weight	Linear
model.language_model.layers.24	Gemma4TextDecoderLayer
model.language_model.layers.24.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.24.mlp	Gemma4TextMLP
model.language_model.layers.24.mlp.down_proj.weight	Linear
model.language_model.layers.24.mlp.gate_proj.weight	Linear
model.language_model.layers.24.mlp.up_proj.weight	Linear
model.language_model.layers.24.per_layer_input_gate.weight	Linear
model.language_model.layers.24.per_layer_projection.weight	Linear
model.language_model.layers.24.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.24.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.24.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.24.pre_feedforward_layernorm.weight	Gemma4RMSNorm

Layer Path	Type
model.language_model.layers.24.self_attn	Gemma4TextAttention
model.language_model.layers.24.self_attn.o_proj.weight	Linear
model.language_model.layers.24.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.24.self_attn.q_proj.weight	Linear
model.language_model.layers.25	Gemma4TextDecoderLayer
model.language_model.layers.25.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.25.mlp	Gemma4TextMLP
model.language_model.layers.25.mlp.down_proj.weight	Linear
model.language_model.layers.25.mlp.gate_proj.weight	Linear
model.language_model.layers.25.mlp.up_proj.weight	Linear
model.language_model.layers.25.per_layer_input_gate.weight	Linear
model.language_model.layers.25.per_layer_projection.weight	Linear
model.language_model.layers.25.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.25.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.25.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.25.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.25.self_attn	Gemma4TextAttention
model.language_model.layers.25.self_attn.o_proj.weight	Linear
model.language_model.layers.25.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.25.self_attn.q_proj.weight	Linear
model.language_model.layers.26	Gemma4TextDecoderLayer
model.language_model.layers.26.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.26.mlp	Gemma4TextMLP
model.language_model.layers.26.mlp.down_proj.weight	Linear
model.language_model.layers.26.mlp.gate_proj.weight	Linear
model.language_model.layers.26.mlp.up_proj.weight	Linear
model.language_model.layers.26.per_layer_input_gate.weight	Linear
model.language_model.layers.26.per_layer_projection.weight	Linear
model.language_model.layers.26.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.26.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.26.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.26.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.26.self_attn	Gemma4TextAttention
model.language_model.layers.26.self_attn.o_proj.weight	Linear
model.language_model.layers.26.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.26.self_attn.q_proj.weight	Linear
model.language_model.layers.27	Gemma4TextDecoderLayer
model.language_model.layers.27.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.27.mlp	Gemma4TextMLP
model.language_model.layers.27.mlp.down_proj.weight	Linear
model.language_model.layers.27.mlp.gate_proj.weight	Linear
model.language_model.layers.27.mlp.up_proj.weight	Linear
model.language_model.layers.27.per_layer_input_gate.weight	Linear
model.language_model.layers.27.per_layer_projection.weight	Linear
model.language_model.layers.27.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.27.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.27.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.27.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.27.self_attn	Gemma4TextAttention
model.language_model.layers.27.self_attn.o_proj.weight	Linear
model.language_model.layers.27.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.27.self_attn.q_proj.weight	Linear
model.language_model.layers.28	Gemma4TextDecoderLayer
model.language_model.layers.28.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.28.mlp	Gemma4TextMLP
model.language_model.layers.28.mlp.down_proj.weight	Linear
model.language_model.layers.28.mlp.gate_proj.weight	Linear
model.language_model.layers.28.mlp.up_proj.weight	Linear

Layer Path	Type
model.language_model.layers.28.per_layer_input_gate.weight	Linear
model.language_model.layers.28.per_layer_projection.weight	Linear
model.language_model.layers.28.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.28.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.28.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.28.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.28.self_attn	Gemma4TextAttention
model.language_model.layers.28.self_attn.o_proj.weight	Linear
model.language_model.layers.28.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.28.self_attn.q_proj.weight	Linear
model.language_model.layers.29	Gemma4TextDecoderLayer
model.language_model.layers.29.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.29.mlp	Gemma4TextMLP
model.language_model.layers.29.mlp.down_proj.weight	Linear
model.language_model.layers.29.mlp.gate_proj.weight	Linear
model.language_model.layers.29.mlp.up_proj.weight	Linear
model.language_model.layers.29.per_layer_input_gate.weight	Linear
model.language_model.layers.29.per_layer_projection.weight	Linear
model.language_model.layers.29.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.29.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.29.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.29.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.29.self_attn	Gemma4TextAttention
model.language_model.layers.29.self_attn.o_proj.weight	Linear
model.language_model.layers.29.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.29.self_attn.q_proj.weight	Linear
model.language_model.layers.30	Gemma4TextDecoderLayer
model.language_model.layers.30.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.30.mlp	Gemma4TextMLP
model.language_model.layers.30.mlp.down_proj.weight	Linear
model.language_model.layers.30.mlp.gate_proj.weight	Linear
model.language_model.layers.30.mlp.up_proj.weight	Linear
model.language_model.layers.30.per_layer_input_gate.weight	Linear
model.language_model.layers.30.per_layer_projection.weight	Linear
model.language_model.layers.30.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.30.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.30.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.30.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.30.self_attn	Gemma4TextAttention
model.language_model.layers.30.self_attn.o_proj.weight	Linear
model.language_model.layers.30.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.30.self_attn.q_proj.weight	Linear
model.language_model.layers.31	Gemma4TextDecoderLayer
model.language_model.layers.31.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.31.mlp	Gemma4TextMLP
model.language_model.layers.31.mlp.down_proj.weight	Linear
model.language_model.layers.31.mlp.gate_proj.weight	Linear
model.language_model.layers.31.mlp.up_proj.weight	Linear
model.language_model.layers.31.per_layer_input_gate.weight	Linear
model.language_model.layers.31.per_layer_projection.weight	Linear
model.language_model.layers.31.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.31.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.31.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.31.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.31.self_attn	Gemma4TextAttention
model.language_model.layers.31.self_attn.o_proj.weight	Linear
model.language_model.layers.31.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.31.self_attn.q_proj.weight	Linear

Layer Path	Type
model.language_model.layers.32	Gemma4TextDecoderLayer
model.language_model.layers.32.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.32.mlp	Gemma4TextMLP
model.language_model.layers.32.mlp.down_proj.weight	Linear
model.language_model.layers.32.mlp.gate_proj.weight	Linear
model.language_model.layers.32.mlp.up_proj.weight	Linear
model.language_model.layers.32.per_layer_input_gate.weight	Linear
model.language_model.layers.32.per_layer_projection.weight	Linear
model.language_model.layers.32.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.32.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.32.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.32.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.32.self_attn	Gemma4TextAttention
model.language_model.layers.32.self_attn.o_proj.weight	Linear
model.language_model.layers.32.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.32.self_attn.q_proj.weight	Linear
model.language_model.layers.33	Gemma4TextDecoderLayer
model.language_model.layers.33.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.33.mlp	Gemma4TextMLP
model.language_model.layers.33.mlp.down_proj.weight	Linear
model.language_model.layers.33.mlp.gate_proj.weight	Linear
model.language_model.layers.33.mlp.up_proj.weight	Linear
model.language_model.layers.33.per_layer_input_gate.weight	Linear
model.language_model.layers.33.per_layer_projection.weight	Linear
model.language_model.layers.33.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.33.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.33.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.33.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.33.self_attn	Gemma4TextAttention
model.language_model.layers.33.self_attn.o_proj.weight	Linear
model.language_model.layers.33.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.33.self_attn.q_proj.weight	Linear
model.language_model.layers.34	Gemma4TextDecoderLayer
model.language_model.layers.34.input_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.34.mlp	Gemma4TextMLP
model.language_model.layers.34.mlp.down_proj.weight	Linear
model.language_model.layers.34.mlp.gate_proj.weight	Linear
model.language_model.layers.34.mlp.up_proj.weight	Linear
model.language_model.layers.34.per_layer_input_gate.weight	Linear
model.language_model.layers.34.per_layer_projection.weight	Linear
model.language_model.layers.34.post_attention_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.34.post_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.34.post_per_layer_input_norm.weight	Gemma4RMSNorm
model.language_model.layers.34.pre_feedforward_layernorm.weight	Gemma4RMSNorm
model.language_model.layers.34.self_attn	Gemma4TextAttention
model.language_model.layers.34.self_attn.o_proj.weight	Linear
model.language_model.layers.34.self_attn.q_norm.weight	Gemma4RMSNorm
model.language_model.layers.34.self_attn.q_proj.weight	Linear
lm_head.weight	Linear
model	Gemma4Model
model.audio_tower	Gemma4AudioModel
model.audio_tower.layers	ModuleList
model.audio_tower.output_proj.weight	Linear
model.audio_tower.output_proj.bias	Linear
model.audio_tower.subsample_conv_projection	Gemma4AudioSubSampleConvProjection
model.audio_tower.subsample_conv_projection.input_proj_linear.weight	Linear
model.audio_tower.subsample_conv_projection.layer0	Gemma4AudioSubSampleConvProjectionLayer
model.audio_tower.subsample_conv_projection.layer0.conv.weight	Conv2d

Layer Path	Type
model.audio_tower.subsample_conv_projection.layer0.norm.weight	LayerNorm
model.audio_tower.subsample_conv_projection.layer1	Gemma4AudioSubSampleConvProjectionLayer
model.audio_tower.subsample_conv_projection.layer1.conv.weight	Conv2d
model.audio_tower.subsample_conv_projection.layer1.norm.weight	LayerNorm
model.embed_audio	Gemma4MultimodalEmbedder
model.embed_audio.embedding_projection.weight	Linear
model.embed_vision	Gemma4MultimodalEmbedder
model.embed_vision.embedding_projection.weight	Linear
model.language_model	Gemma4TextModel
model.language_model.embed_tokens.weight	Gemma4TextScaledWordEmbedding
model.language_model.embed_tokens_per_layer.weight	Gemma4TextScaledWordEmbedding
model.language_model.layers	ModuleList
model.language_model.norm.weight	Gemma4RMSNorm
model.language_model.per_layer_model_projection.weight	Linear
model.language_model.per_layer_projection_norm.weight	Gemma4RMSNorm
model.vision_tower	Gemma4VisionModel
model.vision_tower.encoder	Gemma4VisionEncoder
model.vision_tower.encoder.layers	ModuleList
model.vision_tower.patch_embedder.position_embedding_table	Gemma4VisionPatchEmbedder
model.vision_tower.patch_embedder.input_proj.weight	Linear

4 Parameter Count Estimation

Component	Parameters	% of Total
Embedding	2,751,463,424	53.9%
MLP	1,670,381,568	32.7%
LM Head	402,653,184	7.9%
Attention	379,079,168	7.4%
Other	302,942,848	5.9%
LayerNorm	430,496	0.0%
Total	**5,104,297,504**	**100.0%**

4.1 Per-Layer Parameter Count

Layer	Params	Running Total
model.audio_tower.layers.0.feed_forward1.ffw_layer_1.linear	4,194,304	4,194,304
model.audio_tower.layers.0.feed_forward1.ffw_layer_2.linear	4,194,304	8,388,608
model.audio_tower.layers.0.feed_forward1.post_layer_norm	1,024	8,389,632
model.audio_tower.layers.0.feed_forward1.pre_layer_norm	1,024	8,390,656
model.audio_tower.layers.0.feed_forward2.ffw_layer_1.linear	4,194,304	12,584,960
model.audio_tower.layers.0.feed_forward2.ffw_layer_2.linear	4,194,304	16,779,264
model.audio_tower.layers.0.feed_forward2.post_layer_norm	1,024	16,780,288
model.audio_tower.layers.0.feed_forward2.pre_layer_norm	1,024	16,781,312
model.audio_tower.layers.0.lconv1d.conv_norm	1,024	16,782,336
model.audio_tower.layers.0.lconv1d.depthwise_conv1d	5,120	16,787,456
model.audio_tower.layers.0.lconv1d.linear_end.linear	1,048,576	17,836,032
model.audio_tower.layers.0.lconv1d.linear_start.linear	2,097,152	19,933,184
model.audio_tower.layers.0.lconv1d.pre_layer_norm	1,024	19,934,208
model.audio_tower.layers.0.norm_out	1,024	19,935,232
model.audio_tower.layers.0.norm_post_attn	1,024	19,936,256
model.audio_tower.layers.0.norm_pre_attn	1,024	19,937,280
model.audio_tower.layers.0.self_attn	128	19,937,408
model.audio_tower.layers.0.self_attn.k_proj.linear	1,048,576	20,985,984
model.audio_tower.layers.0.self_attn.post.linear	1,048,576	22,034,560
model.audio_tower.layers.0.self_attn.q_proj.linear	1,048,576	23,083,136
model.audio_tower.layers.0.self_attn.relative_k_proj	1,048,576	24,131,712
model.audio_tower.layers.0.self_attn.v_proj.linear	1,048,576	25,180,288
model.language_model.layers.0.input_layernorm	1,536	25,181,824
model.language_model.layers.0.mlp.down_proj	9,437,184	34,619,008
model.language_model.layers.0.mlp.gate_proj	9,437,184	44,056,192
model.language_model.layers.0.mlp.up_proj	9,437,184	53,493,376
model.language_model.layers.0.per_layer_input_gate	393,216	53,886,592
model.language_model.layers.0.per_layer_projection	393,216	54,279,808
model.language_model.layers.0.post_attention_layernorm	1,536	54,281,344
model.language_model.layers.0.post_feedforward_layernorm	1,536	54,282,880
model.language_model.layers.0.post_per_layer_input_norm	1,536	54,284,416
model.language_model.layers.0.pre_feedforward_layernorm	1,536	54,285,952
model.language_model.layers.0.self_attn.k_norm	256	54,286,208
model.language_model.layers.0.self_attn.k_proj	393,216	54,679,424
model.language_model.layers.0.self_attn.o_proj	3,145,728	57,825,152
model.language_model.layers.0.self_attn.q_norm	256	57,825,408
model.language_model.layers.0.self_attn.q_proj	3,145,728	60,971,136
model.language_model.layers.0.self_attn.v_proj	393,216	61,364,352
model.vision_tower.encoder.layers.0.input_layernorm	768	61,365,120
model.vision_tower.encoder.layers.0.mlp.down_proj.linear	2,359,296	63,724,416
model.vision_tower.encoder.layers.0.mlp.gate_proj.linear	2,359,296	66,083,712
model.vision_tower.encoder.layers.0.mlp.up_proj.linear	2,359,296	68,443,008
model.vision_tower.encoder.layers.0.post_attention_layernorm	768	68,443,776
model.vision_tower.encoder.layers.0.post_feedforward_layernorm	768	68,444,544
model.vision_tower.encoder.layers.0.pre_feedforward_layernorm	768	68,445,312
model.vision_tower.encoder.layers.0.self_attn.k_norm	64	68,445,376
model.vision_tower.encoder.layers.0.self_attn.k_proj.linear	589,824	69,035,200
model.vision_tower.encoder.layers.0.self_attn.o_proj.linear	589,824	69,625,024
model.vision_tower.encoder.layers.0.self_attn.q_norm	64	69,625,088
model.vision_tower.encoder.layers.0.self_attn.q_proj.linear	589,824	70,214,912
model.vision_tower.encoder.layers.0.self_attn.v_proj.linear	589,824	70,804,736
model.audio_tower.layers.1.feed_forward1.ffw_layer_1.linear	4,194,304	74,999,040
model.audio_tower.layers.1.feed_forward1.ffw_layer_2.linear	4,194,304	79,193,344
model.audio_tower.layers.1.feed_forward1.post_layer_norm	1,024	79,194,368
model.audio_tower.layers.1.feed_forward1.pre_layer_norm	1,024	79,195,392
model.audio_tower.layers.1.feed_forward2.ffw_layer_1.linear	4,194,304	83,389,696
model.audio_tower.layers.1.feed_forward2.ffw_layer_2.linear	4,194,304	87,584,000
model.audio_tower.layers.1.feed_forward2.post_layer_norm	1,024	87,585,024
model.audio_tower.layers.1.feed_forward2.pre_layer_norm	1,024	87,586,048
model.audio_tower.layers.1.lconv1d.conv_norm	1,024	87,587,072
model.audio_tower.layers.1.lconv1d.depthwise_conv1d	5,120	87,592,192
model.audio_tower.layers.1.lconv1d.linear_end.linear	1,048,576	88,640,768